

Dataflow Optimization within High-Level Synthesis

Milestone 3: Critical Evaluation and Transfer

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Strengths

- **Speed + quality together:** $\approx 4\times$ faster than flat synthesis while producing circuits of comparable quality. No prior hierarchical method achieved this combination.
- **Non-isomorphic DFG sharing:** the Merge move allows two structurally different DFGs to share one RTL module. All prior methods required separate modules – a significant area overhead this paper eliminates.
- **Global slack propagation:** using the hierarchy’s global view to relax constraints at lower levels enables power savings invisible when optimizing one block alone.
- **Validated at layout level:** the $6.7\times$ power reduction is measured after actual placement and routing, not just estimated – stronger than most HLS papers.

Limitations and Risks

- **Data-flow only:** designs with control flow (**if/else**, variable loops) are completely out of scope. Most real embedded applications mix computation with control.
- **NP-hard module merging:** the RTL embedding problem is at least as hard as graph isomorphism. The heuristic may miss the optimal merging; quality gap is unknown for large modules.
- **Convergence not proven:** a cycle between Merge and Split moves is theoretically possible. No formal termination proof is given.
- **Approximate power model:** the SCM model ignores glitch power and detailed interconnect. Some optimization decisions may be wrong when measured accurately at layout level.

Assumptions That May Be Unrealistic

- **Fixed sampling period:** in adaptive systems the throughput requirement changes at runtime. The method has no mechanism for variable deadlines.
- **Typical input statistics:** the power model uses typical input traces. Worst-case inputs could produce very different switching activity, making power estimates optimistic.
- **User-provided functional equivalence:** the Replace move relies partly on the user declaring which DFGs are equivalent. In practice this information is rarely available in structured form.

Where the Approach is Suitable vs. Unsuitable

Table 1: Applicability to Embedded Systems Scenarios

Suitable	Unsuitable
DSP chips (FIR, IIR, DCT, FFT)	Control-flow intensive firmware
Image/video processing ASICs	FPGA targets (different resource model)
Battery-powered IoT sensor nodes	Systems with variable deadlines
Automotive radar signal processing	Safety-critical systems needing formal optimality guarantees
Large hierarchical designs with tight turnaround requirements	Designs with unavailable IP cores (no Resynthesis possible)

Possible Extensions and Open Questions

- **Control-flow support:** the paper itself identifies this as the primary future direction. Would require control-data flow graph (CDFG) support and new slack-distribution methods.
- **FPGA adaptation:** replacing the standard-cell model with an LUT-based resource model would make the framework applicable to modern FPGA targets.
- **Automatic MAX_MOVES tuning:** a self-adaptive stopping criterion would remove manual parameter tuning.
- **Open question – convergence proof:** it is not proven that the iterative improvement always terminates with a locally optimal solution. A formal analysis would strengthen the theoretical foundation.

References

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- [2] A. Raghunathan and N.K. Jha, “An Iterative Improvement Algorithm for Low Power Datapath Synthesis,” in *Proc. Int. Conf. Comput.-Aided Des.*, Nov. 1995, pp. 597–602.
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